WORKED SOLUTION

STEP III 1992, Question 5

Pure • Vector Geometry in 3D

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q5

The key idea

The symmetry does almost all the work. Reflecting in the plane $y+z=2$ maps the tetrahedron to itself and fixes A and D, so it must swap B and C. That tells us BC is perpendicular to the plane and that its midpoint N lies in the plane. After that, everything is Pythagoras and the scalar product, using the fact that all four vertices are the same distance from the centre Y.

Solution

Finding N and the radius

We have $\overrightarrow{AY} = (3,-2,4) - (0,4,-2) = (3,-6,6)$, so the radius of the sphere is

$$YA = \sqrt{9+36+36} = 9.$$

Since $\overrightarrow{AY} = 3\overrightarrow{YN}$, we get $\overrightarrow{YN} = (1,-2,2)$, so $N = (4,-4,6)$ and $YN = 3$.

Notice that A, Y and N all satisfy $y+z=2$: all three lie in the plane of symmetry.

Why BC is perpendicular to the plane

Let Π be the plane $y+z=2$. Reflection in Π maps the tetrahedron onto itself, so it permutes the vertices. It fixes A and D, which lie in Π. Neither B nor C lies in Π: if one did, the reflection would fix it, and then it would have to fix the fourth vertex too, putting all four vertices in Π and leaving no tetrahedron. So the reflection swaps B and C, which means C is the mirror image of B. Hence

- BC is perpendicular to Π, so it is parallel to the normal $(0,1,1)$;
- the midpoint N of BC lies in Π (which matches what we found).

Ask yourself Why is the triangle YNB right-angled at N?

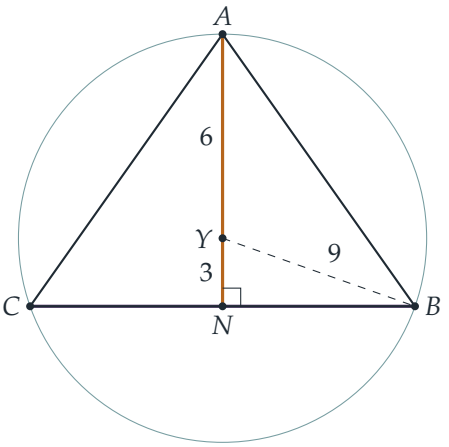
Both Y and N lie in Π, so YN is a line in Π. But NB is along the normal to Π. A normal is perpendicular to every line in the plane, so $\angle YNB = 90°$.

BN and the coordinates of B and C

B lies on the sphere, so $YB = 9$. By Pythagoras in triangle YNB,

$$BN^2 = YB^2 - YN^2 = 81 - 9 = 72, \qquad \text{so} \qquad BN = 6\sqrt{2},$$

as required.



The figure shows the plane ABC, which contains Y (since Y is on AN), drawn to scale. The amber line AN is where it meets Π; the circle is where it cuts the sphere.

A unit vector along BC is $\frac{1}{\sqrt{2}}(0,1,1)$, so

$$\overrightarrow{NB} = \pm 6\sqrt{2} \cdot \tfrac{1}{\sqrt{2}}(0,1,1) = \pm(0,6,6).$$

With $N = (4,-4,6)$, the two ends of the chord are $(4,2,12)$ and $(4,-10,0)$. So

$$B = (4,2,12),\ C = (4,-10,0) \qquad \text{or} \qquad B = (4,-10,0),\ C = (4,2,12).$$

As a check, $(4,2,12) - (3,-2,4) = (1,4,8)$ has length $\sqrt{1+16+64} = 9$, so B really is on the sphere.

The coordinates of D

Write $\overrightarrow{YD} = (p,q,r)$. We have three conditions.

- D and Y both lie in Π, so $\overrightarrow{YD}$ is perpendicular to the normal $(0,1,1)$: $q + r = 0$.
- D is on the sphere: $p^2 + q^2 + r^2 = 81$.
- The angle condition: $\overrightarrow{YA} \cdot \overrightarrow{YD} = YA \cdot YD \cos \angle AYD = 9 \cdot 9 \cdot \frac{1}{3} = 27$.

Ask yourself Three equations in three unknowns, but one is quadratic. How many solutions should we expect?

Up to two, since we will end with a quadratic. Geometrically, D lies on the circle where Π cuts the sphere (a great circle, as $Y \in \Pi$), and the points on that circle making a given angle with YA are the two mirror-image positions either side of A.

With $\overrightarrow{YA} = (-3, 6, -6)$ and $r = -q$, the angle condition gives

$$-3p + 6q + 6q = 27 \quad \Longrightarrow \quad p = 4q - 9.$$

Substituting into the sphere equation, with $r^2 = q^2$:

$$\begin{aligned}(4q-9)^2 + 2q^2 &= 81\\ 18q^2 - 72q &= 0\\ 18q(q-4) &= 0.\end{aligned}$$

- $q = 0$: then $p = -9$, $r = 0$, so $\overrightarrow{YD} = (-9, 0, 0)$ and $D = (-6, -2, 4)$.
- $q = 4$: then $p = 7$, $r = -4$, so $\overrightarrow{YD} = (7, 4, -4)$ and $D = (10, 2, 0)$.

Both give genuine tetrahedra: D is in Π but not on the line AN (the only part of the plane ABC inside Π), so D is not in the plane ABC. Hence

$$D = (-6, -2, 4) \qquad \text{or} \qquad D = (10, 2, 0).$$

Checks

- Each D satisfies $y + z = 2$, and $|\overrightarrow{YD}|^2 = 81$ in both cases (81 and $49 + 16 + 16$).
- For $D = (10, 2, 0)$: $\overrightarrow{YA} \cdot \overrightarrow{YD} = -21 + 24 + 24 = 27 = 81 \cdot \frac{1}{3}$. ✓
- B and C are mirror images: reflecting (x, y, z) in $y + z = 2$ gives $(x, 2 - z, 2 - y)$, which sends $(4, 2, 12)$ to $(4, -10, 0)$.
- A computer searched every integer point of Π with coordinates up to 20 for points on the sphere with $\cos \angle AYD = \frac{1}{3}$, and found exactly $(-6, -2, 4)$ and $(10, 2, 0)$. The tetrahedron has volume $1728/6 = 288$ in both cases, so it is not flat.

Common slips

- Getting N wrong. $\overrightarrow{AY} = 3\overrightarrow{YN}$ puts Y three-quarters of the way along AN, so $N = Y + \frac{1}{3}\overrightarrow{AY}$, not $A + \frac{1}{3}\overrightarrow{AY}$.
- Using the radius as YN or AN. The radius is $YA = 9$; $YN = 3$ is just the distance from the centre to the chord BC.
- Forgetting that D must lie in Π. Without $q + r = 0$ there is a whole circle of possible points.

Beyond STEP

The argument that the reflection must swap B and C is a small piece of group theory: the symmetries of a shape act by permuting its vertices, and fixed points of a reflection must lie in its mirror. The reflection itself is the linear-algebra map $\mathbf{x} \mapsto \mathbf{x} - 2\frac{(\mathbf{x}\cdot\mathbf{n}-d)}{|\mathbf{n}|^2}\mathbf{n}$, an orthogonal transformation with determinant -1. Strang's *Introduction to Linear Algebra* treats reflections and orthogonal matrices, and Allenby's *Rings, Fields and Groups* develops symmetry groups.

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